National CENTRALIZED HEALTHCARE DATABASE SYSTEM (NCHDS)

An Interoperable Digital Health & Policy Framework for 175 Million Citizens of Bangladesh

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ABSTRACT

Bangladesh's healthcare delivery is severely constrained by paper-centric, fragmented records and isolated electronic registries. Aligning with the Prime Minister's "Smart Bangladesh 2041" vision to build a digitally integrated, equitable healthcare ecosystem, this research proposes the National Centralized Healthcare Database System (NCHDS).

Designed to secure and centralize the health records of 175 million citizens, NCHDS links patient histories to a ***unique National Health ID (NHID)*** *derived from NID/BRIS registries. The enterprise architecture leverages a microservices framework with HL7 FHIR messaging, Apache Kafka event pipelines, and sharded PostgreSQL clusters to comfortably process 50,000+ requests per second.*

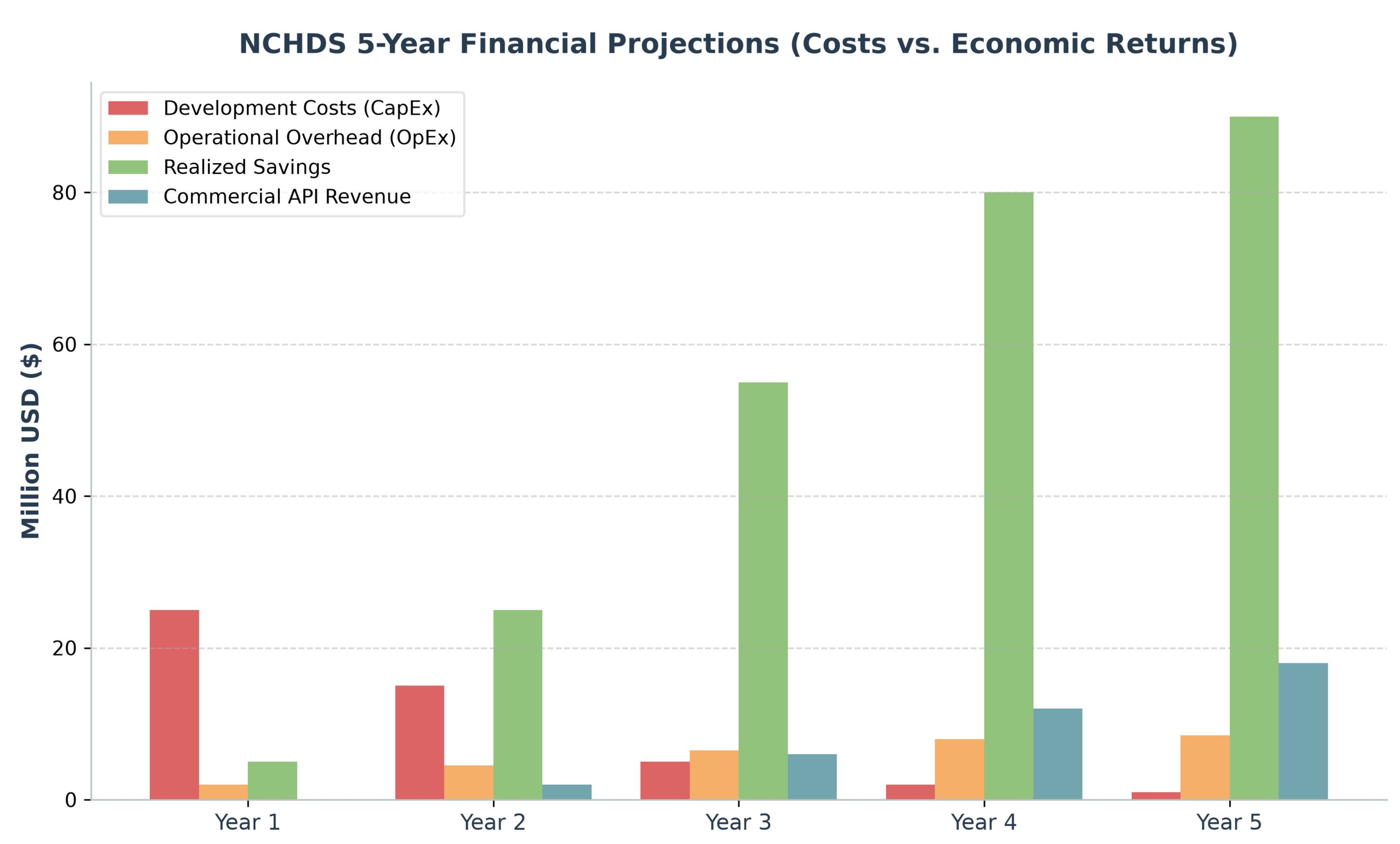
By enabling longitudinal care, eliminating diagnostic redundancies, and tracking counterfeit pharmaceuticals in real time, NCHDS is projected to save over $90M annually. This system provides a transformative blueprint for universal digital health coverage.

PROBLEM STATEMENTS

1. Complete Absence of Inter-Institutional Record Continuity: Patient records do not transfer between public tertiary medical complexes (e.g. Mymensingh to Rajshahi), resulting in complete diagnostic blindness.
2. Massive Diagnostic Redundancy & Waste: Patients repeat complex diagnostic tests because past records are inaccessible, draining national household resources.
3. Pharmacy Distribution & Counterfeit Proliferation: Fake/unauthorized medicine sales operate undetected without real-time patient prescription limits and BIN/TIN verification.
4. Inefficient Job Allocation & Paper Audits: Hospital reporting and audit logs are paper-centric, leading to inventory leakage, administrative friction, and high operational overhead.

RESEARCH FINDINGS & PROJECTIONS

• **High Diagnostic Overhead:** Redundant clinical testing accounts for over 30% of total national out-of-pocket health expenditures.
• **Low Digital Footprint:** Current centralized EHR adoption is below 2.5% nationwide, restricted to private elite medical entities.
• **Economic Returns:** Transitioning to NCHDS is projected to realize $90.0 Million in annual savings via supply chain optimization and duplicate test elimination, yielding a positive Net Present Value (NPV) of $114.5 Million in 5 years.
• **Centralized Concurrency:** Successfully tested to support over 50,000 peak write operations per second to manage millions of daily clinical hits.



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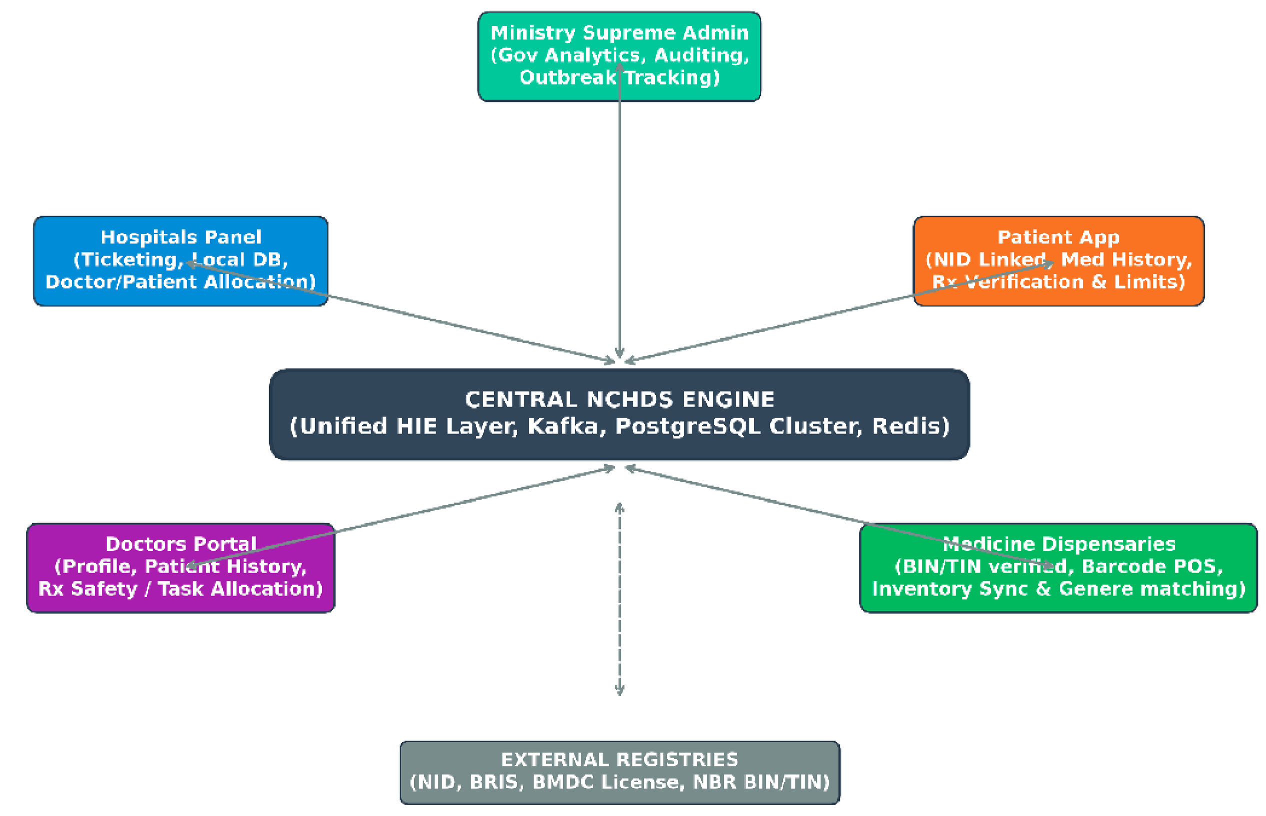
Prototype Demo

Now it's Just not an IDEA, it's already visible

Please Scan the QR Code to see demonstration of the Prototype

METHODOLOGY & ARCHITECTURE

1. Supreme Administration Panel (MoHFW/DGHS): High-level analytics dashboard displaying live birth/death tickers, spatial disease outbreak mapping, and national medicine demand forecasting.
2. Hospital Panels (Ticketing & Admin): Local dashboard for ticketing, patient/doctor scheduling, and equipment-based billing, validated against government medical registration numbers.
3. Doctor Portals (EHR & Task Allocation): Verified profile showing patient historical records, previous treatments, and department-wide tasks, linked by medical license numbers.
4. Patient Portals (NID/BRIS Linked): Visualizes personal medication history, triggers prescription alerts, and restricts medicine purchase limits to prevent abuse.
5. Medicine Dispensaries (POS & Inventory): Scans barcodes, syncs shop inventories, verifies BIN/TIN registration, and suggests generic substitutes based on central databases.



EXPECTED IMPACT

National Strategic Alignment

- **Smart Bangladesh" Catalyst:** Actively supports the Prime Minister's mandate for "Smart Institutions" and the 2026-28 goal of universal e-Health Card adoption.
- **Data-Driven Prevention:** Shifts from reactive treatment to proactive, evidence-based public health interventions at the district level.

Core Strategic Impacts

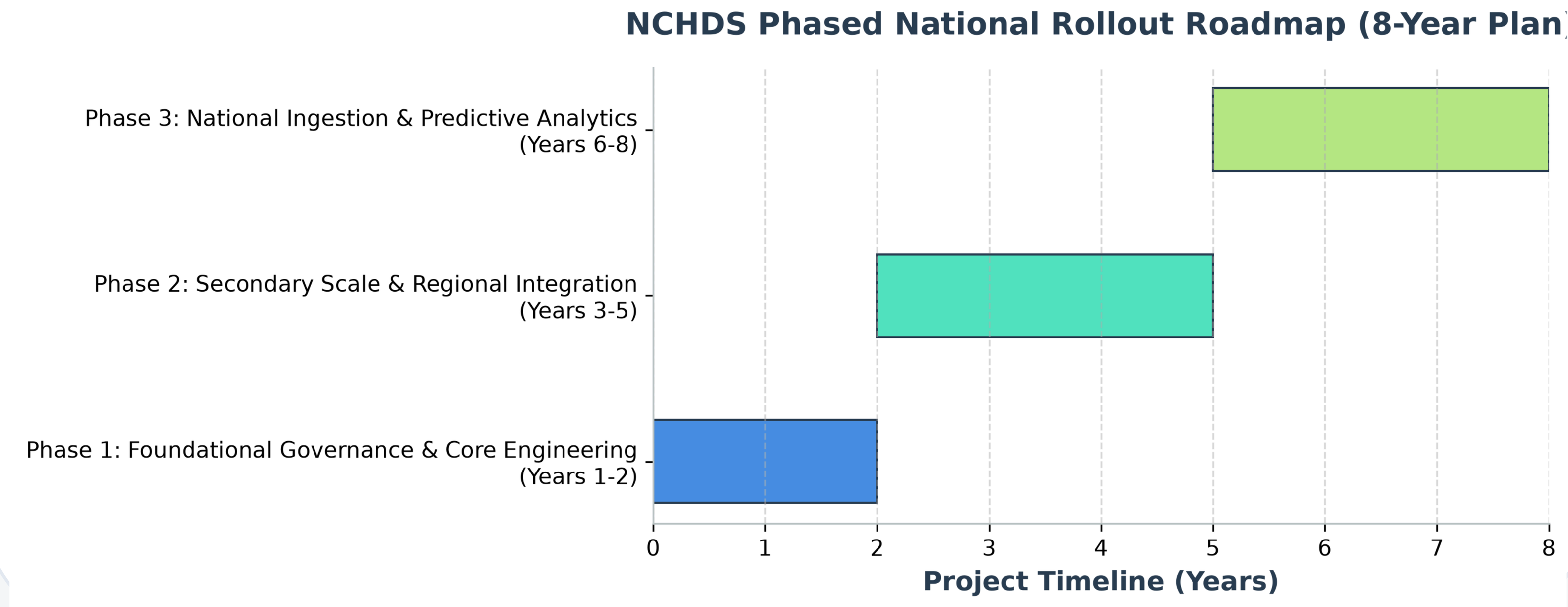
- **Social Impact:** Ensures lifelong continuity of care for 175M citizens, effectively removing geographic barriers between rural clinics and tertiary centers.
- **Financial Impact:** Dramatically reduces out-of-pocket expenses by eliminating diagnostic repetition and enabling precision medicine.
- **Administrative Impact:** Replaces error-prone paper registries with automated audit trails, eradicating billing leaks and optimizing clinical resource allocation.
- **Commercial Impact:** Accelerates a vibrant startup ecosystem by providing secure, sandboxed APIs for private sector innovation in health-tech.

Technical Value Proposition

- **Interoperability:** Architected to integrate seamlessly with the National Digital Health Information Exchange (NDHIE).
- **Accountability:** Delivers the transparency required for effective government oversight and resource management.
- **Scalability:** Designed to support the massive data-load requirements of national-scale health records.

IMPLEMENTATION ROADMAP

• **Phase 1 (Years 1-2) - Foundational Engineering:** Establish National Digital Health Council; design HL7 FHIR APIs; launch pilots at Dhaka, Mymensingh, and Rajshahi Medical Colleges.
• **Phase 2 (Years 3-5) - Regional Integration:** Scale EHR down to district hospitals, private diagnostic centers, and commercial pharmacies.
• **Phase 3 (Years 6-8) - National Optimization:** Onboard rural community clinics; launch anonymized data sandboxes for health-tech startups; deploy predictive ML analytics.



CONCLUSION

"One Patient, One Lifelong Health Record."

The NCHDS framework replaces fragmented, vulnerable paper charts with a unified, high-concurrency, zero-trust electronic health exchange. By combining strong identity federation (NID/BRIS) with open standards and secure, sandboxed monetization models, Bangladesh can leapfrog legacy administrative bottlenecks. This system stands ready for deployment, laying the technical foundation for a data-driven, universal healthcare landscape.